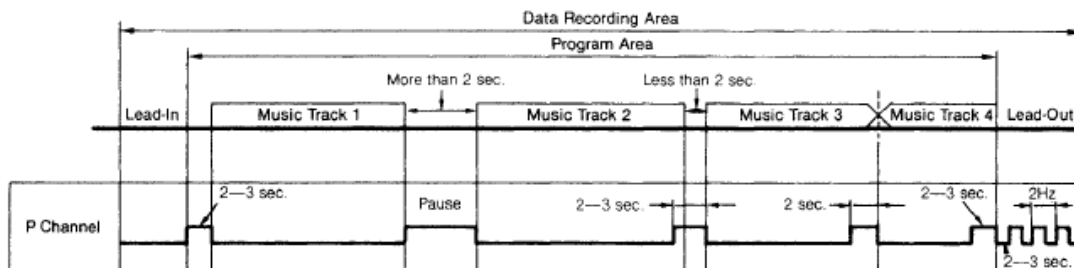


5.1

1. The datarate of the subcode in subcode block/s is 75 Hz. A subcoding block should consists of 98 subcoding symbols which each have a size of 8 bits, but the first 2 subcoding symbols are replaced by the subcoding sync patterns S0 and S1. This means the subcode has a datarate of 59600 bits/s ($75 * 96 * 8$). (BS_Standard 17.3)

2.



(BS_Standard page 80)

3. This information is stored in the Q-channel which is one of the subcoding channels. As mentioned in question 1 a subcoding block consists of 98 subcoding symbols. This means a subcoding block contains 98 Q-bits. The first 2 Q-bits are part of the subcoding sync patterns, the next 4 are control bits and the 4 bits after that are the address bits that determine the mode which tells what information the following 72 bits will hold. Only if the address bits determine that the mode is mode 1, these 72 bits will contain info about the number of tracks, the starting positions of the tracks and the total playing time of the audio CD. The first 8 bits of the 72-bit datafield will tell you which track the subcoding block is part of. If these bits are 00 it means the block is part of the lead-in track which means it will contain additional information about the real tracks. Blocks that are part of the lead-in track will have the fields POINT, PMIN, PSEC and PFRAME. If the value of POINT is A0 in a block PMIN will give the track number of the first track on the disc, if POINT has value A1 PMIN will give the track number of the last track on the disc. This way the total number of tracks can be deducted. If the value of POINT is between 01 and 99 the fields PMIN, PSEC and PFRAME will contain the absolute time at which the track with the same track number as the value in POINT will start. If POINT has value A2 PMIN, PSEC and PRFAME point to the absolute time the lead-out track starts. Since you now know when the first track starts and the lead-out track starts, you also know the total playing time of the audio.

To protect the data from errors 2 different techniques are used. First of all, during the duration of the lead-in track this info about the timing of the tracks is continuously looped and is also always repeated 3 times in subsequent subcoding blocks. Secondly the last 16 Q-bits

of every subcoding block are part of a Cyclic Redundancy Check code, which is a checksum which helps to detect errors. (BS_Standard 17.5)

5.3

1. In the encoder the delay of 2 frames means that if a block with even and odd samples arrives as input a delay of 2 symbols is placed between the even and odd samples. This means that all the even samples in the block are delayed with 2 blocks. Because of this, if 2 subsequent uncorrectable blocks occur, interpolation can still be accomplished to estimate the values. Next an interleaving sequence happens. This means the symbols are scrambled so as to separate even- and odd-numbered data words. If an error now happens concealment can still be performed, which is essentially covering up the error. After this the C2 encoder accepts the 24-byte parallel word and produces 4 bytes of Q parity. Q parity is designed to correct one erroneous symbol or up to four erasures in one word. Now its time for the delay lines of unequal length. Here the 28 bytes are delayed by differing periods, which are integer multiples of 4 blocks. As a result, each C2 word is stored in 28 different blocks, distributed over 109 blocks. Next the C1 encoder accepts a 28-byte word (from 28 different C2 words) and produces 4 more bytes of P parity. The C1 encoder is used to correct single symbol errors and to detect and flag double and triple errors for Q correction. Lastly the delay of 1 frame introduces a fixed delay of one symbol to alternate symbols. This odd/even delay spreads the output words over two data blocks. This prevents random errors from disrupting more than one symbol in one word, even if two adjacent symbols in one block are erroneous.

The operations in the decoder reverse these operations.

C1 code is designed to correct random errors and detect burst errors. Short-duration random errors can still be corrected but longer burst errors are detected, flagged and passed along. With the info from the C1 decoder and after de-interleaving the C2 decoder is able to correct the longer burst errors and also the residual random errors. (Ken C. p69-71)

3. The C2 decoder can correct up to 4 erasures. This means that to have an uncorrectable error with minimum distance, the burst should span 5 symbols long when it arrives at the C2 decoder. Since the encoder displaces symbols with 4 blocks after C2 encoding, a burst of 17 blocks would affect 5 symbols but a burst of 16 blocks would only affect 4 symbols. So a burst of 16 blocks can always be corrected by the C2 decoder. Blocks after encoding contain 32 symbols of 8 bits. This means that the maximum duration of a burst can be 16 blocks x 32 symbols/block x 8 bits/symbol, which is 4096 bits. 1 Frame/block contains 6 stereo samples and because the audio sample rate is 44100 samples/s this means the frame rate is 7350 frames/s. This means the maximum duration a burst can occur is 16 frames / 7350 frames/s, which is 0.00218 seconds. Because the scanning velocity of the laser is 1.3 m/s, this means the maximum width of a scratch is 1.3m/s * 0.00218 s, which is 0.00283 meter or 2.83 millimeter.

4. The interpolator works because in the real world sound-waves are also smooth. A soundwave can't jump from one position to another. If you use linear interpolation to draw a line between the safe samples, this will also result in a smooth curve which is a lot better than abruptly stopping the wave.

Because the sample frequency is 44.1 kHz the error of 8 samples only has a duration of 8 samples / 44100 samples/s, which is 0.00018 seconds or 0.18 milliseconds. Since audible audio frequencies only go up to about 20 kHz, the physical sound wave is limited in how fast it can oscillate. This means the time of 0.18 milliseconds is way too short for a physical sound wave to change a lot, so the linear line will be a good approximation. (Ken C. p19, p65-67)

6. EFM stands for eight-to-fourteen modulation and translates blocks of 8 databits into blocks of 14 bits using a ROM dictionary. The 14-bit patterns are chosen to control the number of transitions between 0 and 1 so a greater data density can be achieved. Since 256 different code patterns are chosen out of the 16384 possible combinations, code patterns will now differ by more than one bit from other code patterns, whereas for the 8-bit patterns flipping one bit would always result in a different code pattern. EFM thus provides a kind of error detection. The 14-bit combinations that are chosen have more than 2 but less than 10 0s appear continuously, which ensures bits can be read correctly on the CD. The blocks of 14 bits are linked with 3 merging bits.

As said the 14-bit patterns have 16384 possible combinations but only 256 of them are valid codewords. If the EFM demodulator receives one of the illegal patterns it can signal to the CIRC an error has occurred and because the CIRC now knows the location of the error it will have to spend less time correcting the error. (Ken C. p79-80)